PANSKURA BANAMALI COLLEGE (AUTONOMOUS)

BCA 2nd Semester Examination, 2022

PAPER - BCACC - 4T

[*Discrete Mathematics*]

Full Marks : 60

Time : 3 hours

(*The figure in the margin indicate full marks. Candidates are required to give their answers in their own words as far as practicable)*

Answer any 04 questions : **$4 \times 15 = 60$**

1. (a)Without using truth tables prove that $(\sim p \vee q) \wedge (p \wedge (p \wedge q)) \equiv p \wedge q$.

(b) Prove that for every positive integer n, $n^3 + n$ is even.

(c) Define Tautalogy and Contradiction. Verify that the following proposition is a Tautalogy

$$((p \vee q) \wedge (p \to r) \wedge (q \to r)) \to r.$$

(d) Using Mathematical induction prove that $6^{n+2} + 7^{2n+1}$ is divisible by 43 for all $n \in N$. $2\frac{1}{2} + 2\frac{1}{2} + (2+3) + 5$

2. (a) For any three arbitrary sets A, B, C Prove that $A - (B \cup C) = (A - B) \cap (A - C)$ and $(A \cap B) \cap (A - B) = \phi$.

(b) If R be a relation in the set of integers Z defined by $R = \{(x, y) : x, y \in Z, (x - y)$ is divisible by $5\}$. Prove that R is an equivalence relation.

(c) Define bijective mapping with an example. State the reason whether the mapping $f : [-\pi, \pi] \to [-1, 1]$ defined by $f(x) = \sin x \ \forall \ x \in [-\pi, \pi]$ is bijective or not.

$5 + 5 + (2 + 3)$

3. (a) If $f : A \to B$ and $g : B \to C$ be one to one and onto function, then prove that $g \circ f$ is also one to one, onto function and $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.

(b) Define symmetric difference between two sets with an example. Use $k-$ map to simplify the Boolean function

$$a'b'c' + a'b'c + a'bc + a'bc' + ab'c + abc.$$

(c) Using Mathematical induction prove that

$$\frac{1}{1\cdot 2}+\frac{1}{2\cdot 3}+\frac{1}{3\cdot 4}+\cdots+\frac{1}{n(n+1)}=\frac{n}{n+1}$$

$5+(2+3)+5$

4. (a) Prove that the complete disjunctive normal form and complete conjunctive normal form in three variables in a Boolean algebra is equal to the unit and zero element of the Boolean algebra.

(b) Solve the recurrence relation $a_r - 3a_{r-1} + 3a_{r-2} - a_{r-3} = 0$ and given that $a_0 = 2, a_1 = 1, a_2 = 1$

(c) A certain computer centre employs 100 computer programmers. Of these 47 can program in FORTRAN, 35 in PASCAL, 20 in COBOL and 23 can program in FORTRAN and PASCAL, 12 in COBOL and FORTRAN, 11 in PASCAL and COBOL and 5 in FORTRAN, PASCAL and COBOL. How many can program in none of these 3 languages ? $5+5+5$

5. (a) Draw the Hasse diagram representing the partial ordering $\{(A,B) : A \subseteq B\}$ on the power set P(S), where $S = \{a, b, c\}$. Find the maximal, minimal, greatest and least elements of the poset.

(b) Prove that in a group $(G, \circ)$ for all $(a, b) \in G$ each of the equations $a \circ x = b$ and $y \circ a = b$ has unique solution in G

(c) Solve the recurrence relation using generating function $a_r - 5a_{r-1} + 6a_{r-2} = 2^r$, given that $a_0 = 1, a_1 = 1$. $5+5+5$

6. (a) Give an example of a relation which is

(i) reflexive and transitive but not symmetric

(ii) reflexive and symmetric but not transitive

(iii) an equivalence relation.

(b) Define linear recurrence relation with constant coefficient.

(c) Let Q be the set of rational numbers and operation $'\star'$ is defined by $a \star b = a + b + ab, \quad \forall \quad a, b \in Q$. Examine whether $(Q, \star)$ is a group or not.

(d) Define generating function. Find the generating function for the sequence $1, 2, 2^2, 2^3, \cdots$.

$3+2+5+(2+3)$